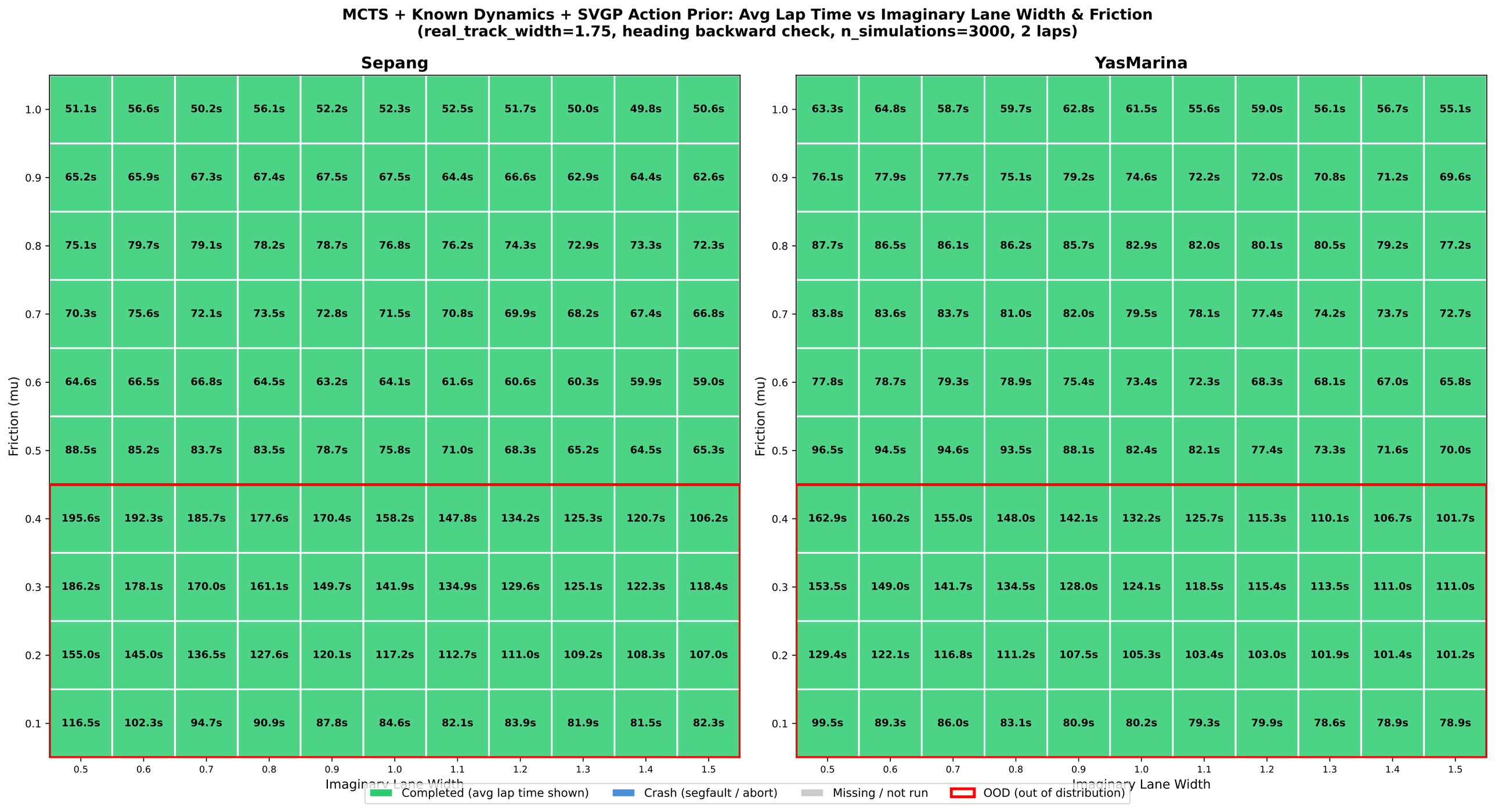
MCTS + Known Dynamics + SVGP Action Prior: Avg Lap Time vs Imaginary Lane Width & Friction
(real_track_width=1.75, heading backward check, n_simulations=3000, 2 laps)
Sepang
YasMarina
Friction (mu)
Imaginary Lane Width
1.0 51.1s 56.6s 50.2s 56.1s 52.2s 52.3s 52.5s 51.7s 50.0s 49.8s 50.6s
0.9 65.2s 65.9s 67.3s 67.4s 67.5s 67.5s 64.4s 66.6s 62.9s 64.4s 62.6s
0.8 75.1s 79.7s 79.1s 78.2s 78.7s 76.8s 76.2s 74.3s 72.9s 73.3s 72.3s
0.7 70.3s 75.6s 72.1s 73.5s 72.8s 71.5s 70.8s 69.9s 68.2s 67.4s 66.8s
0.6 64.6s 66.5s 66.8s 64.5s 63.2s 64.1s 61.6s 60.6s 60.3s 59.9s 59.0s
0.5 88.5s 85.2s 83.7s 83.5s 78.7s 75.8s 71.0s 68.3s 65.2s 64.5s 65.3s
0.4 195.6s 192.3s 185.7s 177.6s 170.4s 158.2s 147.8s 134.2s 125.3s 120.7s 106.2s
0.3 186.2s 178.1s 170.0s 161.1s 149.7s 141.9s 134.9s 129.6s 125.1s 122.3s 118.4s
0.2 155.0s 145.0s 136.5s 127.6s 120.1s 117.2s 112.7s 111.0s 109.2s 108.3s 107.0s
0.1 116.5s 102.3s 94.7s 90.9s 87.8s 84.6s 82.1s 83.9s 81.9s 81.5s 82.3s
0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5
1.0 63.3s 64.8s 58.7s 59.7s 62.8s 61.5s 55.6s 59.0s 56.1s 56.7s 55.1s
0.9 76.1s 77.9s 77.7s 75.1s 79.2s 74.6s 72.2s 72.0s 70.8s 71.2s 69.6s
0.8 87.7s 86.5s 86.1s 86.2s 85.7s 82.9s 82.0s 80.1s 80.5s 79.2s 77.2s
0.7 83.8s 83.6s 83.7s 81.0s 82.0s 79.5s 78.1s 77.4s 74.2s 73.7s 72.7s
0.6 77.8s 78.7s 79.3s 78.9s 75.4s 73.4s 72.3s 68.3s 68.1s 67.0s 65.8s
0.5 96.5s 94.5s 94.6s 93.5s 88.1s 82.4s 82.1s 77.4s 73.3s 71.6s 70.0s
0.4 162.9s 160.2s 155.0s 148.0s 142.1s 132.2s 125.7s 115.3s 110.1s 106.7s 101.7s
0.3 153.5s 149.0s 141.7s 134.5s 128.0s 124.1s 118.5s 115.4s 113.5s 111.0s 111.0s
0.2 129.4s 122.1s 116.8s 111.2s 107.5s 105.3s 103.4s 103.0s 101.9s 101.4s 101.2s
0.1 99.5s 89.3s 86.0s 83.1s 80.9s 80.2s 79.3s 79.9s 78.6s 78.9s 78.9s
0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5
Completed (avg lap time shown)
Crash (segfault / abort)
Missing / not run
OOD (out of distribution)